
BlindTasks: A Verifiable Vision Gym for Reinforcement Learning

Low-Level Visual Reasoning in Vision-Language Models

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Abstract

Despite achieving near-human performance on complex semantic and visual QA benchmarks, modern vision-language models still struggle with elementary geometric and spatial primitives. Recent analyses reveal that these failures originate from the decoding layer rather than the visual encoder, as internal visual features contain sufficient information to predict the ground truth. In this work, we investigate whether *reinforcement learning with verifiable rewards* (RLVR) can resolve this readout failure.

Taking inspiration from the *BlindTest* benchmark, we introduce *BlindTasks*: a training environment featuring tasks of varying difficulty across elementary cognitive primitives, such as counting and determining orientation.

Training APERTUS-8B and QWEN2.5-VL-7B in this gym, we provide a systematic study of RLVR behavior along three key dimensions. First, we show that a decoder-only update is sufficient to improve the model’s pass@1 score on these primitives. Second, we investigate the role of data composition by designing task-balanced, primitive-balanced, and difficulty-driven training distributions, mapping how these profiles affect both in-distribution and by-task performance. Additionally, we evaluate the broader impact of this training by evaluating both on related and general-purpose benchmarks. Third, we analyze the performance of base versus post-trained models at higher values of k . This allows us to study whether the known RLVR trade-off (improving pass@1 while failing to increase, or even diminishing, pass@ k) applies in this context, or if we are genuinely augmenting the models’ capabilities. By exploring how this trade-off evolves at varying temperatures, we also present a novel finding regarding how training data composition affects format compliance as inference temperature increases.

1 Introduction

Despite scoring above 90 % on complex semantic benchmarks like document-VQA and chart-VQA, modern vision language models (VLMs) fail on elementary tasks that humans solve trivially. [Rahmanzadehgervi et al. \[2025\]](#) recently demonstrated that state-of-the-art VLMs struggle with basic spatial perception, achieving a mean accuracy of only 58.07 % on rudimentary tasks like counting intersecting lines or resolving overlapping circles. Crucially, linear probes on the vision encoders of these models reach a score $\geq 99.47\%$, indicating that the bottleneck is not a lack of visual information, but rather a failure of the language head to effectively leverage the information contained in the visual embeddings.

We attribute this failure to four interacting layers:

1. **Encoder objective mismatch:** CLIP-style pre-training focuses on semantic similarity rather than the geometric topology required to distinguish if two lines actually touch or merely pass near each other. Consequently, the language model struggles to extract task-relevant signals from the visual embeddings.
2. **Visual-token bottleneck:** The image is compressed into a few hundred tokens where each token covers a patch of roughly 14×14 to 32×32 pixels. If two circles overlap within that space, the discriminating feature may be smaller than the resolution of a single token.
3. **Late-fusion architecture:** Because the image is encoded once before the question is processed, the model cannot re-examine the raw image to refine its representation. It is constrained by a fixed set of tokens that may not have prioritized the specific geometric features required by the prompt.
4. **Language prior dominance:** When visual tokens are ambiguous, the language model’s internal priors take over. For instance, a model may predict "5" for any logo resembling the Olympic rings, regardless of how many rings are actually rendered in the image. This phenomenon was highlighted in [Vo et al. \[2026\]](#).

In this work, we investigate whether reinforcement learning with verifiable rewards (RLVR) can address this failure by specifically targeting late-fusion constraints (3) and language prior dominance (4). We introduce *BlindTasks*, a training gym featuring procedurally-generated tasks across 10 visual primitives, grouped for analysis into 6 cognitive families (Appendix A.1), such as counting, detecting orientation and identifying overlapping shapes. Our intervention is guided by three core hypotheses:

- **H1: A decoder-only update is sufficient.** Given that the encoder representations already contain sufficient information for near-perfect scores, updating only the projector and language head should recover the missing performance.
- **H2: Group-relative advantage provides a contrastive signal.** By comparing multiple rollouts for the same image, RL directly penalizes prior-dominated, incorrect answers.
- **H3: Per-instance variation forces grounding.** By randomizing shape and background colors, stroke size, and other spurious details, we prevent the model from exploiting canvas-specific shortcuts, forcing it instead to rely on true perceptual primitives.

We provide a systematic study of these hypotheses across APERTUS-8B and QWEN2.5-VL-7B, evaluating the impact of data composition, the pass@1 versus pass@k trade-off, and cross-benchmark transfer.

2 Background and Related Work

Modern open-weight VLMs follow a late-fusion recipe where a vision encoder produces prompt-independent representations that are mapped via a projector into the LLM’s embedding space [Liu et al. \[2023\]](#). Because the image is encoded only once before the question is processed, the visual representation is fixed. In this study, we focus on APERTUS-8B and QWEN2.5-VL-7B. While both sit in the late-fusion family, they differ in their visual processing pipelines: Apertus utilizes a discrete IBQ tokenizer, whereas Qwen employs a Vision Transformer paired with an MLP projection layer.

RLVR has emerged as a dominant strategy for eliciting reasoning in LLMs. We utilize Group Relative Policy Optimization (GRPO) [Shao et al. \[2024\]](#), which eliminates the need for a separate critic model by computing advantages relative to a group of rollouts generated for the same prompt. While models like DeepSeek-R1 have demonstrated the efficacy of this approach in text-based reasoning, recent extensions (such as Visual-RFT and R1-V) have only recently begun applying similar techniques to the multimodal domain.

Concurrently, recent work has questioned whether RLVR genuinely improves reasoning capabilities or primarily reshapes the distribution of generated outputs. [Yue et al. \[2025\]](#) observed that RL can substantially improve pass@1 while simultaneously reducing pass@k, suggesting a trade-off between solution quality and output diversity. While this phenomenon is well-documented in language-based reasoning, it remains unclear if a similar behavior emerges in visual reasoning settings.

We ground our investigation of these dynamics in the visual shortcomings identified by the BlindTest suite [Rahmanzadehgervi et al. \[2025\]](#). While related benchmarks like Eyes-Wide-Shut [Tong et al. \[2024\]](#) and CountBench [Paiss et al. \[2023\]](#) have also exposed VLM deficits in counting and spatial relationships, they remain static evaluation datasets. *BlindTasks* bridges this gap by transforming these insights into a procedural training environment. This enables the first systematic study of RL-driven perceptual grounding, allowing us to evaluate whether the pass@1 versus pass@k trade-off emerges in low-level visual tasks and whether training benefits transfer to related out-of-distribution benchmarks.

3 Methodology

3.1 BlindTasks Gym

BlindTasks is a procedurally generated visual reasoning gym designed for RLVR. The environment is derived from the tasks introduced in BlindTest [Rahmanzadehgervi et al. \[2025\]](#) and extended with custom tasks targeting counting, spatial reasoning, orientation estimation, and metric comparison. Example images of the tasks and the primitive groups are shown in Appendix A.1.

Procedural Generation. To discourage memorization and shortcut learning, we randomize visual attributes that are irrelevant to the underlying reasoning task, including stroke thickness, fill and background colors, typography, and object placement. To prevent data leakage, we partition the seed ranges used for the training, validation, and test split generations. We also craft multiple prompt variations for each task to prevent overfitting to a specific linguistic template. Templates are standardized per-task to specify the description, valid answer set, and required answer format; details and examples are provided in Appendix A.2.



Figure 1: Five seeds of `olympic.interlocking` showing per-seed `RenderConfig` variation.

The objective is to ensure that successful policies must solve the underlying reasoning problem rather than exploit spurious visual correlations. Furthermore, we define both an easy and a hard variant for every task. Hard variants increase visual clutter, overlap, ambiguity, or distractor density while preserving the underlying reasoning primitive.

Dataset composition We generate a train split of 22000 samples under three different construction strategies:

- **Uniform:** Equal allocation across tasks (1 k rows per task). Because counting accounts for 13 of the 22 task variants, this strategy implicitly allocates $\sim 59\%$ of training rows to it.

- **Cognitive:** Allocation is first balanced across primitives (~ 3.7 k rows per primitive) and then equally distributed within tasks of the same family. Counting receives $\sim 28\%$ of rows (roughly half its allocation under the uniform strategy), while other primitives receive roughly double.
- **Inverse-baseline:** Allocation is proportional to task difficulty as measured by baseline pass@1 performance. Specifically, $w_t \propto \max(0.05, 1 - \text{pass}@1_t)$. The floor of 0.05 prevents any task from receiving zero rows. Here, counting-related tasks represent $\sim 66\%$ of the rows.

The validation and test split are consistently generated using the uniform composition, containing 500 and 4400 samples, respectively.

3.2 Reinforcement Learning Training

Training is implemented using the *verl* framework Sheng et al. [2024]. We fine-tune models using GRPO Shao et al. [2024]: for every prompt, we sample $G = 8$ candidate completions and compute relative advantages using the GRPO estimator.

Rewards are provided by a deterministic verifier, which examines the generation, extracts the answer from the required format (e.g., `\boxed{\cdot}` for Qwen), and assigns a binary reward through exact string matching with the ground truth.

Unless otherwise stated, all experiments use a learning rate of 10^{-6} , a batch size of 64, a GRPO group size of 8, and a KL penalty coefficient of 10^{-3} . Responses are truncated to a maximum length of 512 tokens.

Checkpoint selection We measure validation performance on held-out tasks every 10 optimizer steps. If the pass@1 score fails to improve for three consecutive evaluation steps, we stop the training and select the checkpoint with the highest validation performance.

3.3 Evaluation Protocol

Unless stated, generations are sampled using temperature $T = 0.6$, top- $p = 0.95$, and a maximum generation length of 16384 tokens. Later sections study the effect of alternative sampling temperatures. We evaluate our checkpoints on both our in-distribution test split and on the following out-of-distribution benchmarks: CountBench Paiss et al. [2023], BlindTest Rahmanzadehgervi et al. [2025], MMBench Liu et al. [2024], and POPE Li et al. [2023].

For evaluation on BlindTasks, we sample $n = 128$ completions per prompt and report pass@ k metrics. Pass@128 is measured directly, while pass@ k for $k < 128$ is estimated using the unbiased estimator introduced by Chen et al. [2021]. In addition to aggregate scores, we report per-task performance, per-primitive aggregations, and held-out primitive transfer results. For CountBench, we use the same evaluation protocol as for BlindTasks and report pass@1 through pass@128. For BlindTest, MMBench, and POPE, we limit generation to $n = 8$ samples per prompt and report pass@1 and pass@8 metrics.

Scoring. We assign a pass or fail score to each generation using deterministic verifiers, following the same approach described in Section 3.2. In the temperature ablation studies (§4.3), we employ a lenient parser (Appendix A.2) to address format compliance degradation.

4 Experiments and results

4.1 Train data composition comparisons

Table 1 reports the pass@ k scores for the three data mixes described in (§3.1). All three runs lift pass@1 from 0.236 to 0.74–0.78, confirming our first hypothesis: a few hundred GRPO optimizer steps on the BlindTasks gym substantially improve per completion accuracy.

Two surprises follow:

Simpson’s paradox. The cross-mix, per-primitive breakdown tells a more detailed story: the inverse-baseline mix achieves the highest performance on counting tasks, the cognitive

Table 1: Evaluation results of best checkpoints for each train composition

Run	best step	pass@1	pass@8	pass@32	pass@128
Apertus base (no RL)	—	0.236	0.605	0.784	0.858
Apertus + uniform	343	0.773	0.810	0.823	0.829
Apertus + cognitive	310	0.744	0.799	0.818	0.828
Apertus + inverse-baseline	320	0.779	0.820	0.834	0.840

mix excels on `letter_id`, `direction`, `spatial`, and `yes_no`, while all three mixes reach saturation on metric tasks. The aggregate ranking therefore depends on the averaging convention. Under micro-averaging (weighted by problem count, dominated by counting’s 13/22 share of the test set) `invbase` wins by 1.1 pp. However, under macro-averaging (equal weight per primitive), `cognitive` wins by 1.5 pp on `pass@1` and 1.9 pp on `pass@128`. This is a *Simpson’s-paradox* effect at the aggregation level: the data mix acts as an allocation knob rather than a universally “best” recipe. Table 7 details the exact cross-over between the micro and macro evaluation regimes. We report both aggregations in Appendix B.

The `pass@k` tradeoff. We observe that the `pass@k` degradation documented in language reasoning tasks also applies here. All three RL runs *lose* `pass@128` relative to the untrained baseline. The model trades sampling breadth for deterministic accuracy. However, as we demonstrate in §4.2, this degradation can be recovered at higher inference temperatures.

Qwen sanity-check. We treat QWEN2.5-VL-7B (Bai et al. [2025]) as a positive control that the pipeline is model-agnostic. While the `pass@1` improvement is less pronounced (with the baseline achieving a `pass@1` score of 0.59 against the best post-trained score of 0.796), the `pass@k` degradation occurs even earlier than in APERTUS, with the base model overtaking the RL-trained variant by `pass@32` (full curves are provided in Appendix E).

4.2 Decoding temperature separates capability from channel failures

The standard interpretation of `pass@k` degradation following RLVR is that probability mass is redistributed toward fewer, more likely correct answers. This naturally suggests that increasing temperature may recover diversity and close the `pass@k` gap. In Figure 2, we evaluate the best inverse-baseline checkpoint across $T \in \{0.6, 0.8, 1.0, 1.2, 1.4, 1.6, 1.8, 2.0\}$. `Pass@128` increases monotonically up to $T = 1.8$, peaking at 0.924. This represents improvements of +8.4 pp and +6.6 pp over the same checkpoint and the base model, evaluated at $T = 0.6$. `Pass@1` is much less sensitive to temperature: it peaks at $T = 1.4$, rising only from 0.779 to 0.791, before slightly declining at higher temperatures. These curves show that the apparent `pass@k` degradation at $T = 0.6$ is largely a sampling artifact: by increasing temperature, we recover diversity without sacrificing single-sample accuracy.

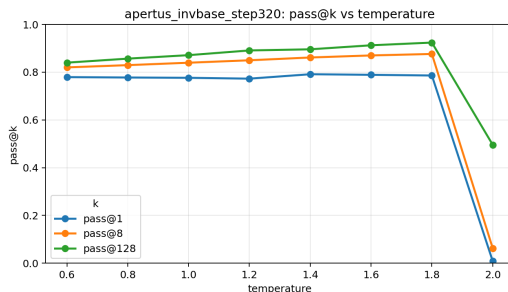


Figure 2: Temperature sweep on the inverse-baseline checkpoint.

Only at $T = 2.0$ does performance collapse: under strict parsing, `pass@1` drops to almost zero with a score of 0.008, while `pass@128` falls to 0.496. Table 2 also shows that the minimum per-task format-validity rate drops to 2.8%. This suggests that the model is no longer reliably preserving the structured output channel required by the verifier, motivating a second evaluation with the *lenient* parser described in §3.3. Under this regime, `pass@1`

remains poor at 0.091, but pass@128 recovers to 0.856, close to the base-model ceiling. Thus, at very high temperature, individual completions are often malformed or non-committal, but correct answers are still present somewhere among the 128 samples.

Table 2 highlights how the high temperature affects the performance drop: only 5.8% of completions are strict-correct, with an additional 5.5% recoverable under the lenient parser. The remaining generations are dominated by pure rambles and wrong answers, confirming that the $T = 2.0$ collapse is not a loss of visual capability but rather a breakdown of the answer channel. Appendix C provides a more granular taxonomy of failure modes.

Table 2: Temperature diagnostics for the inverse-baseline checkpoint.

$T = 2.0$ parse outcomes			Strict temperature diagnostics			
Outcome	Count	%	T	$\Delta_{0.6}$	Δ_{base}	min fmt-valid
Strict-correct: proper wrapper, correct answer	32,678	5.8	0.6	–	–1.8 pp	~ 100%
Lenient-correct: correct answer, missing wrapper	31,076	5.5	1.4	+5.6 pp	+3.8 pp	99.9%
Recoverable but wrong: parseable answer, incorrect	160,164	28.4	1.8	+8.4 pp	+6.6 pp	99.5%
Pure ramble: no parseable commitment within token limit	339,282	~ 60	2.0	–34.4 pp	–36.2 pp	2.8%

4.3 Cross-mix at high temperature

To test whether the temperature findings generalize across mixes, we re-evaluate all three checkpoints and the untrained baseline at $T=1.4$ and $T = 1.8$ with both strict and lenient parsers. Table 3 shows that all data mixes reach $\text{pass@128} \in [0.77, 0.90]$ at $T=1.4$ under lenient scoring, and further improve by going to $T=1.8$: invbase +2.8 pp, cognitive +4.8 pp, uniform +6.6 pp. Only the inverse-baseline mix exceeds the baseline’s pass@128 at high temperatures.

Table 3: Cross-mix evaluation at high temperatures. The Δ measures wrapper failure: cases where an answer is recoverable by the lenient parser but not accepted by the strict verifier.

Checkpoint	pass@128 (lenient)	pass@128 (strict)	Δ_{parsing}	min fmt-valid
$T = 1.4$				
Apertus + invbase 320	0.896	0.896	~ 0	99.9%
Apertus + cognitive 310	0.808	0.482	+33 pp	1.8%
Apertus + uniform 343	0.772	0.559	+21 pp	4.4%
Baseline	0.858	0.647	+21 pp	4.5%
$T = 1.8$				
Apertus + invbase 320	0.924	0.924	~ 0	99.5%
Apertus + cognitive 310	0.856	0.484	+37 pp	2.0%
Apertus + uniform 343	0.838	0.509	+33 pp	3.1%

Data composition affects format compliance. The strict-vs-lenient gap is mix-dependent: the inverse-baseline checkpoint maintains format validity at both temperatures, while cognitive, and uniform show gaps of 21 to 37 percentage points. The baseline has a 21 pp gap of the same kind. For completeness, the pass@1 performance is also comparable between baseline, cognitive and uniform, with all of them scoring <0.02 across temperature values, whereas inverse baseline achieves a strict score of 0.791 at $T=1.4$ and 0.786 at $T=1.8$, as shown in Figure 2. Apart from the data mix, everything is identical across the training runs, ruling out a hyperparameter explanation: the training data mix alone determines structured-output discipline at inference time. A plausible explanation is discussed in §5.

4.4 Generalization and Transfer

We next investigate whether the models generalize beyond the specific tasks seen during training. We evaluate transfer at two levels: internally, by testing on held-out spatial primitives within the BlindTasks gym, and externally, by testing on out-of-distribution benchmarks (CountBench, BlindTest, MMBench, and POPE).

Held-Out Primitive Transfer. To test internal transfer, we train an inverse-baseline model on a subset of 14 of the 22 BlindTasks variants. We then evaluate the resulting checkpoint on the full 4,400-example test set, partitioned into four categories: in-distribution tasks, within-primitive tasks (counting), cross-primitive similar tasks (direction), and cross-primitive dissimilar tasks (letter identification).

Table 4: Held-out primitive transfer performance.

Cell	n tasks	pass@128
In-distribution (trained)	14	0.801 ± 0.015
Within-primitive (counting)	4	0.710 ± 0.030
Cross-primitive similar (direction)	2	0.835 ± 0.036
Cross-primitive dissimilar (letter id)	2	0.863 ± 0.034

As shown in Table 4, transfer performance does not predictably degrade with cognitive distance. Cross-primitive tasks achieve pass@128 scores comparable to (or higher than) the in-distribution baseline (0.801). The only substantial performance drop occurs in the within-primitive cell (0.710). However, this ~ 9.1 pp gap is entirely driven by the grid-hard task, which reaches a pass@128 of just 0.050. Excluding this outlier, the within-counting average rises to 0.900. This indicates that internal capability transfer is limited primarily by visual complexity (e.g., severe clutter or overlap) rather than cognitive category distance.

External Benchmark Transfer. We evaluate the base APERTUS model and its post-trained variants on external benchmarks to probe both low-level perceptual transfer (CountBench, BlindTest) and broader multimodal retention (MMBench, POPE).

Table 5: Pass@1 and Pass@8 scores of base and post-trained APERTUS variants.

Model	Pass@1					Pass@8				
	MMB	POPE	CB	BT	Avg.	MMB	POPE	CB	BT	Avg.
Base	45.4	67.0	54.5	30.9	49.5	78.6	95.1	83.7	71.0	82.1
Invbase	66.3	83.9	85.9	62.8	74.7	78.7	90.6	91.5	77.3	84.5
Cognitive	66.4	84.1	85.3	60.2	74.0	78.3	90.3	91.2	73.7	83.4
Uniform	66.4	83.8	85.8	63.2	74.8	78.1	90.1	91.2	77.7	84.3

As shown in Table 5, all RL-trained models demonstrate consistent pass@1 improvements across the board. The gains are most significant on CountBench and BlindTest, confirming that RLVR does not merely memorize the synthetic gym distribution, but actively improves the model’s generalized visual read-out capabilities.

However, the pass@k trade-off observed in-distribution also transfers to these external datasets. By pass@8, the baseline model catches up—and on MMBench and POPE, often overtakes the post-trained models. This reinforces that while RLVR successfully concentrates probability mass on correct single completions, it sacrifices sampling diversity at larger values of k . Figure 3 illustrates this flattened pass@k trajectory on CountBench.

Finally, we observe that the *uniform* and *cognitive* data mixes perform just as well as the *inverse-baseline* mix on CountBench, despite containing significantly fewer counting examples. This further proves that external transfer is not driven by task-specific heuristics or sheer example volume, but by a fundamental improvement in generalized perceptual grounding.

5 Discussion

The main result is that reinforcement learning substantially improves single-sample accuracy on BlindTasks. Across the three Apertus data mixtures, pass@1 increases from 0.236 to 0.74–0.78, whereas [Rahmanzadehgervi et al. \[2025\]](#) reported SFT collapse at 0.368 on a comparable budget. The Qwen ablation exhibits the same qualitative trend. This supports H1: updating the decoder-side components is sufficient to recover a large fraction of the missing readout performance on these low-level visual primitives. The per-primitive gain pattern is also consistent with H2: the largest pass@1 improvements appear on the most prior-dominated

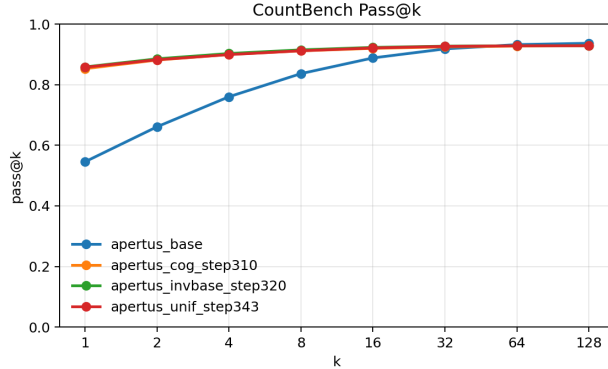


Figure 3: Pass@ k performance of the four APERTUS versions on **CountBench**.

baseline tasks (see Table 8) where the group-relative advantage is sharpest because only the minority of correct rollouts receive positive credit while the prior-dominated majority is penalized in lockstep. However, the results should not be interpreted as showing that RLVR uniformly expands the model’s capability: all post-trained Apertus checkpoints lose pass@128 relative to the base model, showing that the pass@1/pass@ k trade-off previously observed in language reasoning also applies to vision tasks.

The pass@ k trade-off is temperature-dependent. For the inverse-baseline checkpoint, temperature increase recovers pass@128 from 0.840 at $T = 0.6$ to 0.924 at $T = 1.8$, exceeding the base-model pass@128 ceiling. Thus, the default-temperature pass@128 loss is largely a sampling artifact: correct answers remain accessible in the model distribution, but require more diverse sampling to appear among the completions. At very high temperature, however, performance collapses because the model no longer reliably preserves the required format.

Data composition affects format compliance. The most unexpected result is that training data composition controls high-temperature format robustness. Ceteris paribus, the inverse-baseline mixture remains wrapper-compliant up to $T = 1.8$, while the uniform and cognitive mixtures exhibit large performance gaps between strict and lenient parsing already at $T = 1.4$. We hypothesize that this is due to gradient concentration: inverse-baseline oversamples harder tasks, producing mixed GRPO groups for longer and therefore sustaining reward gradients on both answer content and output format.

Transfer is promising but uneven. External benchmarks show broad pass@1 improvements on CountBench, BlindTest, MMBench, and POPE, suggesting that the model is not just memorizing the synthetic training distribution. The strongest transfer appears on CountBench and BlindTest, which are closest to the low-level primitives targeted by the gym. Unexpectedly, we observe the pass@ k trade-off on these broader out-of-distribution benchmarks as well: the base model catches up or outperforms the post-trained models at pass@8 on MMBench and POPE, despite being substantially worse at pass@1. Internal held-out ablations also reveal that cross-primitive transfer can be strong, but achieving a good performance on visually difficult tasks (e.g., grid-hard) requires in-distribution training.

Limitations. Several limitations qualify our conclusions. First, our by-task and by-primitive analysis follows our intuition of which tasks are similar or dissimilar. The model’s internal representation is not necessarily aligned with the human perception of the tasks. This argument likewise applies to our difficulty splits. Second, the format compliance results apply to our specific prompt templates. We did not test alternative templates. Third, the lenient parser is itself a biased evaluation artifact, useful for separating channel failures from capability failures but not a substitute for the strict verifier used during training.

Conclusion. Overall, BlindTasks shows that decoder-side RLVR can strongly improve low-level visual readout, but also that pass@1 alone is insufficient to characterize this improvement. Sampling diversity, format compliance, decoding temperature, and data composition all shape the observed behavior. The gym therefore serves not only as a training environment, but also as a diagnostic tool for separating capability, sampling, and answer-channel failures in RL-trained vision-language models.

6 Future work

The ablations in this report surface several promising new directions, expanding upon our original work plan of algorithmic sweeps and transfer experiments.

Training Dynamics

- **Adaptive inverse-baseline:** Re-estimate per-task pass@1 every N steps and re-weight the train mix online, so the allocation tracks the policy’s *current* capability instead of a frozen pre-RL snapshot. Adaptive weighting is more compute-expensive but would let the wrapper-discipline signal track the actual learning trajectory.
- **Counterfactual stress tests:** Based on *VLMs Are Biased* Vo et al. [2026], we will extend the gym with out-of-distribution (OOD) variants where the visual prior is *wrong* to probe whether RL trains the policy to follow *evidence* or just sharpens the existing visual prior.

Algorithmic and Suite Expansions

- **Algorithm sweep:** Try DAPO Yu et al. [2025], RLOO Ahmadian et al. [2024], and GSPO Zheng et al. [2025] for the actor objective and compare them with GRPO in terms of training stability and convergence speed. We will also study how the resulting models perform at higher pass@ k values versus GRPO, whose performance has been shown to degrade. The same study can be extended to other RLVR algorithms.
- **Task-suite expansion:** Extend the gym with new primitives that probe other failure modes. Some candidates include analog-clock reading, polygon vertex-counting, angle estimation, graph-edge counting, bar-chart value extraction, and symmetry detection.

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A BlindTasks Gym

A.1 Task catalogue

BlindTasks contains 22 procedurally generated task variants. Each variant isolates one of 10 low-level visual primitives, with multiple variants targeting the same primitive at different difficulties. We group the 10 primitives into 6 broader cognitive families for analysis. Table 6 lists the variants by family, and Figure 4 shows representative renders.

Table 6: The 22 BlindTasks variants grouped by cognitive family.

Primitive	Variants
Counting	Nested Squares (easy + hard), Circles (easy), Lines (easy + hard), Olympic Rings (easy + hard), Grid (easy + hard), Subway (easy + hard), Olympic Interlocking, Word Grid
Binary spatial judgment	Circles (hard; whether the two circles touch)
Label identification	Circled Letter (easy + hard)
Geometric orientation	Orientation (easy + hard)
Relative spatial position	Spatial Relation (easy + hard)
Metric magnitude	Size Compare (easy + hard)

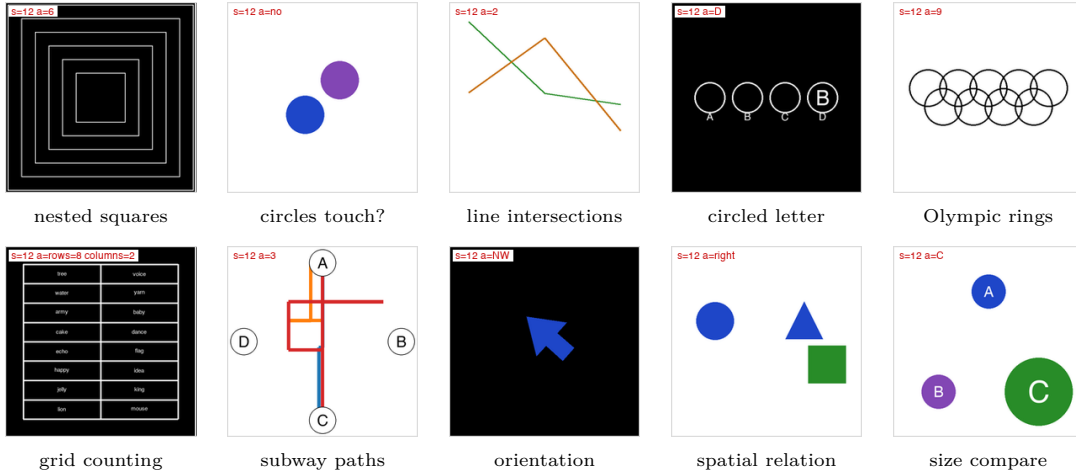


Figure 4: Representative BlindTasks renders. Ten of the 22 variants are shown.

The easy and hard variants preserve the same underlying primitive but alter visual nuisance factors such as clutter, overlap, distractor density, and object spacing. This lets us separate improvements on the intended primitive from overfitting to a particular visual template.

A.2 Generation, prompts, and verification

Procedural generation. Every BlindTasks instance is generated from a seed. The seed determines the image, the natural-language task instruction, and the ground-truth answer. During generation we randomize nuisance factors such as stroke width, foreground and background colors, font, object placement, distractor density, and question phrasing. These factors do not change the correct answer, but they prevent the policy from memorizing a small bank of fixed images.

Per-instance diversity. BlindTest uses a finite collection of rendered images, which is appropriate for evaluation but problematic for RL training: a policy repeatedly seeing the same canvases could memorize configuration tuples rather than learn the perceptual primitive. BlindTasks therefore replaces the fixed image bank with a streamable sampler. Each rollout samples a render configuration, including image size, line width, color scheme, font, object placement, and prompt paraphrase. We render at $2\times$ supersampling and downsample with

LANCZOS to reduce aliasing artifacts. The sampled render configuration is stored with the ground truth for post-hoc analysis.

Train/validation/test split. Train, validation, and test examples use disjoint seed ranges. We maintain a convention of seed ranges:

1. **Train:** Seeds go from 0 up to 1’000’000 non inclusive.
2. **Validation:** Seeds go from 1’000’000 to 2’000’000 non inclusive.
3. **Test:** Seeds go from 2’000’000 to 3’000’000 non inclusive.

When generating datasets, we dedicate a seed to each task instance: we start from the first seed of the range, and progressively increase it by one every time a new row of the dataset is generated.

Prompting strategy. Each task prompt specifies three things: the task objective, the valid answer set, and the required answer channel. Early prompt templates mixed a bare-answer instruction with a boxed-answer instruction. For example, an old size-comparison prompt was:

```
Of the three labelled circles, which has the biggest radius?
Reply with A, B, or C.

Put your final answer in \boxed{ }.
Reply with the single value only (no extra words inside the box).
```

When evaluating the Qwen base version, we realized that this wording caused the model to emit bare answers such as A, which are visually meaningful but fail strict parsing. The reported experiments use a standardized format that makes the answer channel explicit:

```
Of the three labelled circles, which has the biggest radius?
Put your final answer inside \boxed{ } as one of A, B, or C.
For example: \boxed{A}
```

The model may produce intermediate natural-language reasoning, but the verifier scores only the final boxed answer. For instance, a generation ending in `\boxed{A}` is parseable even if the preceding text contains an explanation.

Answer Parsing During our experiments we mostly employ the strict parser, which extracts the answer from a specific format (e.g `\boxed{.}` for Qwen, and from the display-answer tool for apertus). When evaluating Apertus at higher temperatures, we also employ a lenient parser that extends the strict parser by accepting: (a) a bare JSON of the form `{"answers": [...]}`; (b) “the answer is X” natural-language phrasing; (c) a bare leading character or number.

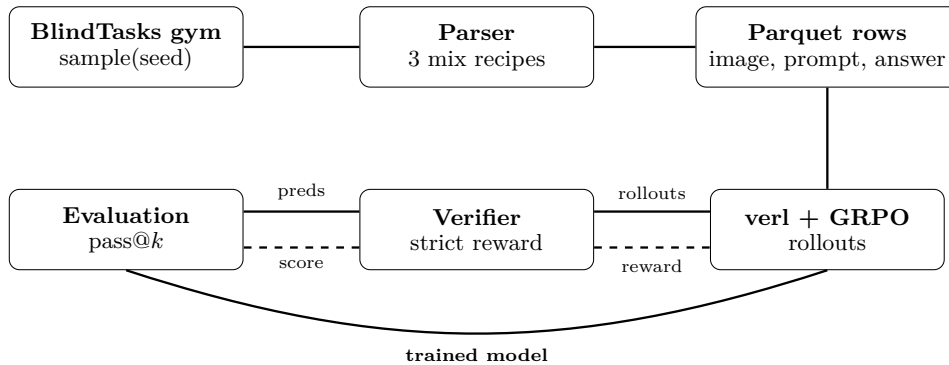


Figure 5: End-to-end BlindTasks pipeline.

B In-distribution ablations

Apertus pass@ k curves. Figure 6 shows the pass@ k curves for the Apertus baseline and the three RL-trained variants at the default decoding temperature. RL sharply improves pass@1, but at $T = 0.6$ all trained variants have lower pass@128 than the base model.

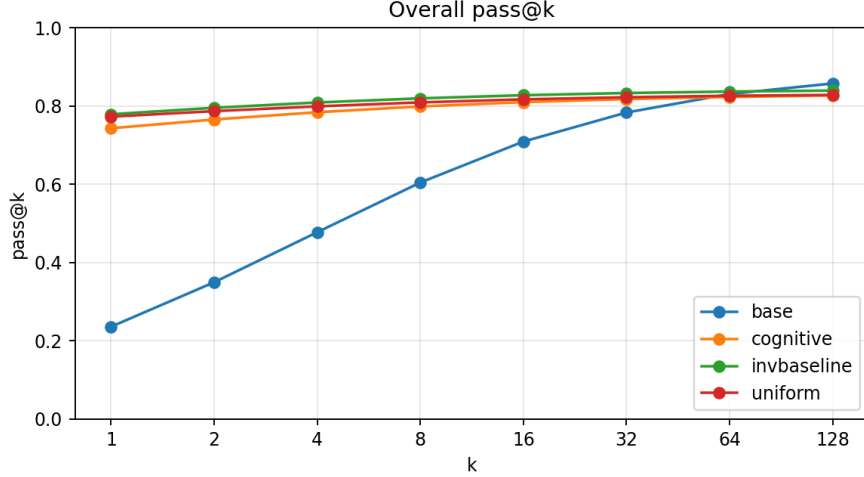


Figure 6: Apertus pass@ k curves at the default decoding temperature ($T = 0.6$, top- $p = 0.95$).

Table 7: Micro- and macro-averaged performance at $T = 0.6$ under strict parsing. Micro-averaging weights all test problems equally; macro-averaging weights primitive families equally.

Run	pass@1 micro	pass@1 macro	pass@128 micro	pass@128 macro
Apertus base	0.236	0.340	0.858	0.876
Apertus + inverse-baseline	0.779	0.851	0.840	0.896
Apertus + uniform	0.773	0.854	0.829	0.896
Apertus + cognitive	0.744	0.875	0.828	0.915

The aggregation choice changes the apparent winner. Counting accounts for 13 of the 22 task variants, so micro-averaging favours the inverse-baseline mixture, which allocates more mass to counting. Macro-averaging gives equal weight to primitive families and favours the cognitive mixture. This is the Simpson’s-paradox effect discussed in the main text.

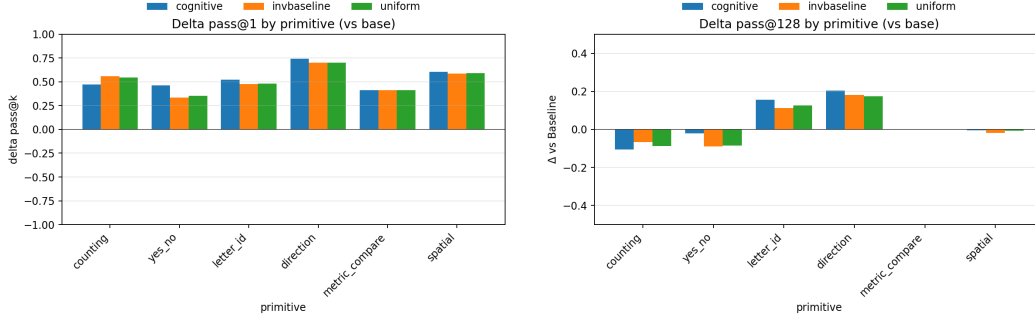
Table 8: Per-primitive pass@1 at $T = 0.6$ under strict parsing.

Primitive	n tasks	Baseline	Invbase	Uniform	Cognitive
Counting	13	0.148	0.707	0.694	0.620
Direction	2	0.221	0.924	0.923	0.962
Letter ID	2	0.189	0.663	0.668	0.713
Metric	2	0.589	1.000	1.000	1.000
Spatial	2	0.384	0.970	0.977	0.988
Yes/no	1	0.509	0.844	0.860	0.970

Largest task-level gains and remaining hard cases. Table 9 reports the five largest pass@1 gains and five lowest post-RL pass@1 tasks. The largest improvements occur on hard counting and orientation tasks, while grid and circled-letter hard variants remain difficult.

B.1 By-task and by-primitive performance

Figure 7 shows the gain per primitives, while Figure 8 contains a more fine-grained view, showing by-task pass@1 performance. We observe that the pass@1 vs pass@128 tradeoff does not apply uniformly across primitives: counting, spatial and binary primitives have a negative delta, whereas performance on direction and letter primitives is still superior to the base model.



(a) pass@1 deltas

(b) pass@128 deltas

Figure 7: Per-primitive changes relative to the Apertus baseline.

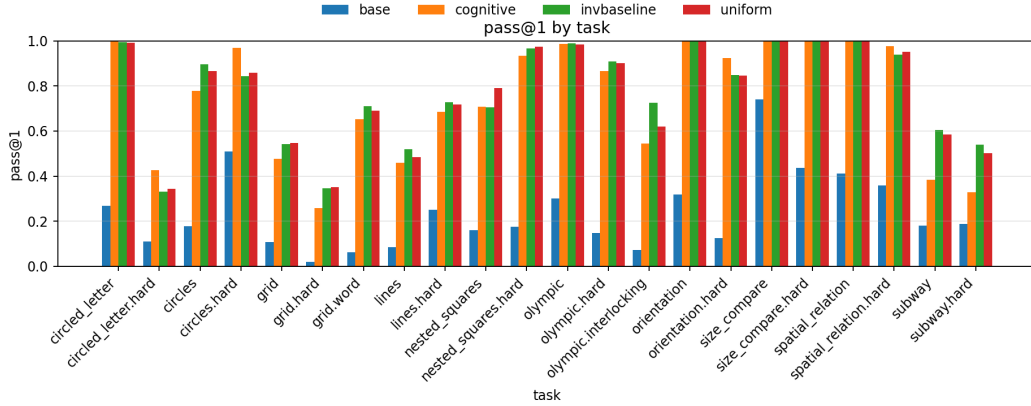


Figure 8: pass@1 performance of the different model versions across all the tasks

We also note that the RL pass is enough to saturate pass@1 performance on tasks where the base model achieves low scores, such as size comparison and spatial relations. In Table 9 we report the tasks where RL achieves the biggest pass@1 gains, using the inverse baseline checkpoint as reference.

B.2 By-difficulty performance

As mentioned in §3.1 we generate tasks with both an easy and a harder variant. In Figure 9 we report the by-difficulty performance of both the base and post-trained Apertus models. Interestingly, we observe that the cross-over happens earlier and more markedly in easier tasks than harder ones.

Table 9: Task-level pass@1 extremes after RL. Gains are absolute percentage-point improvements over the baseline.

Task	post-RL pass@1	gain
<i>Largest gains</i>		
Nested Squares (hard)	0.967	+79.2 pp
Olympic (hard)	0.940	+76.2 pp
Circled Letter	0.995	+72.7 pp
Orientation (hard)	0.907	+72.5 pp
Orientation	1.000	+68.2 pp
<i>Lowest post-RL pass@1</i>		
Grid (hard)	0.351	+33.2 pp
Circled letter (hard)	0.425	+31.6 pp
Lines	0.519	+43.4 pp
Subway (hard)	0.539	+35.1 pp
Subway	0.605	+42.5 pp

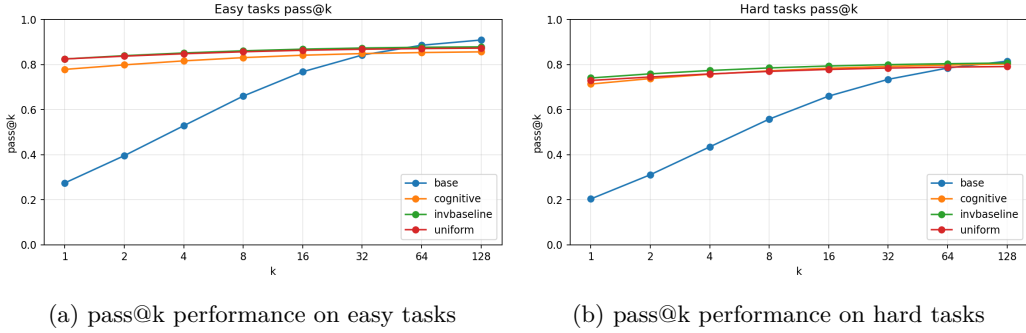


Figure 9: easy vs. hard tasks pass@k performance

C Temperature and channel diagnostics

Temperature sweep. The inverse-baseline checkpoint was evaluated across temperatures $T \in \{0.6, 0.8, \dots, 2.0\}$. Figure 10 shows that pass@128 increases up to $T = 1.8$, recovering the default-temperature pass@k deficit, before collapsing at $T = 2.0$.

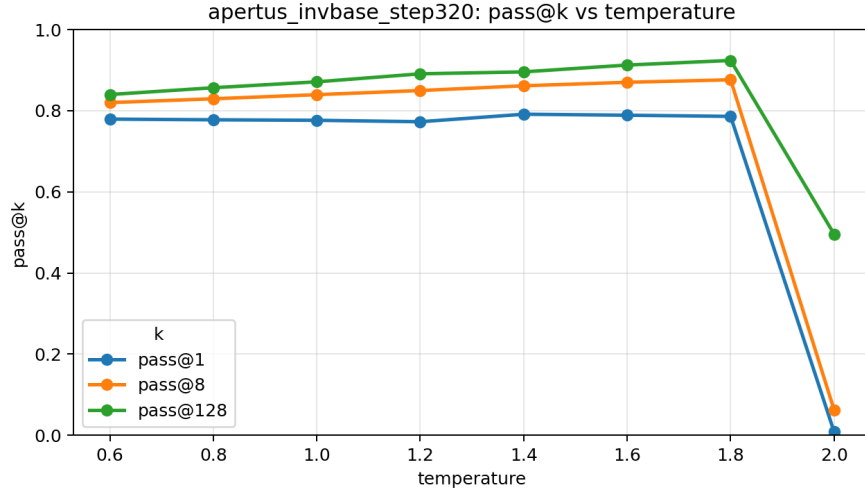


Figure 10: Temperature sweep for the inverse-baseline checkpoint under strict parsing.

Table 10: Selected temperature diagnostics for the inverse-baseline checkpoint. Deltas are absolute pass@128 differences relative to the same checkpoint at $T = 0.6$ and relative to the base-model pass@128 ceiling of 0.858.

Setting	pass@1	pass@128	$\Delta_{0.6}$	Δ_{base}	min fnt-valid
$T = 0.6$ strict	0.779	0.840	—	−1.8 pp	~ 100%
$T = 1.4$ strict	0.791	0.896	+5.6 pp	+3.8 pp	99.9%
$T = 1.8$ strict	0.786	0.924	+8.4 pp	+6.6 pp	99.5%
$T = 2.0$ strict	0.008	0.496	−34.4 pp	−36.2 pp	2.8%
$T = 2.0$ lenient	0.091	0.856	−0.8 pp	−0.2 pp	—

The $T = 2.0$ cliff. At $T = 2.0$, strict performance collapses because generations often stop respecting the answer protocol. Table 11 decomposes 563,200 rollouts into four parse outcomes. The key observation is that strict-correct generations are only a small minority, while many generations are either recoverable only by the lenient parser or contain no parseable commitment at all.

Table 11: Parse outcomes at $T = 2.0$ for the inverse-baseline checkpoint.

Outcome	Count	Proportion
Strict-correct: wrapper present and answer correct	32,678	5.8%
Lenient-correct: correct answer, missing wrapper	31,076	5.5%
Recoverable but wrong: parseable answer, incorrect	160,164	28.4%
Pure ramble: no parseable commitment	339,282	60.3%

Primitive-level effect of the cliff. Table 12 compares the best strict high-temperature point ($T = 1.8$) against lenient $T = 2.0$. Binary yes/no tasks remain robust because the answer space is tiny, while counting and letter identification lose more because they require many valid samples to cover a larger answer space.

Table 12: Primitive-level pass@128 under high-temperature decoding.

Primitive	$T = 1.8$ strict	$T = 2.0$ lenient	Difference
Yes/no	0.945	1.000	+5.5 pp
Spatial	0.993	0.958	−3.5 pp
Direction	0.985	0.930	−5.5 pp
Letter ID	0.905	0.835	−7.0 pp
Counting	0.875	0.798	−7.7 pp

Wrapper discipline across mixes. Table 13 shows that format robustness depends strongly on the training mixture. The inverse-baseline checkpoint remains wrapper-compliant at $T = 1.4$ and $T = 1.8$, while the cognitive and uniform mixtures suffer large strict–lenient gaps.

D Held-out primitive transfer

The held-out experiment trains an inverse-baseline policy on 14 of the 22 task variants and evaluates on all tasks at $T = 1.4$ with lenient parsing. The goal is to probe whether RL learns transferable visual readout behavior or merely task-specific heuristics. We treat this experiment as exploratory because the held-out cells are not matched for baseline difficulty and several tasks are close to saturation.

The within-counting cell is the only cell substantially below in-distribution. This is driven mainly by **grid.hard**, whose held-out pass@128 is only 0.050; excluding it raises the within-counting average to 0.900. Conversely, the letter-ID cell contains only two tasks and includes saturated variants, so the apparent advantage over in-distribution should not be interpreted as evidence that dissimilar primitives transfer better. The conservative conclusion is that cross-primitive transfer can be strong, but the split is confounded by task difficulty.

Table 13: Cross-mix evaluation at high temperature.

Checkpoint	pass@128 lenient	pass@128 strict	gap	worst fmt-valid
$T = 1.4$				
Apertus + inverse-baseline	0.896	0.896	~ 0	99.9%
Apertus + cognitive	0.808	0.482	+33 pp	1.8%
Apertus + uniform	0.772	0.559	+21 pp	4.4%
Apertus base	0.858	0.647	+21 pp	4.5%
$T = 1.8$				
Apertus + inverse-baseline	0.924	—	—	99.5%
Apertus + cognitive	0.856	0.484	+37 pp	2.0%
Apertus + uniform	0.838	0.509	+33 pp	3.1%

Table 14: Held-out transfer-cell pass@128 at $T = 1.4$ with lenient scoring. Confidence intervals are bootstrap 95% CIs over problems with $B = 1000$ resamples.

Cell	n tasks	n problems	pass@128	95% CI
In-distribution (trained)	14	2800	0.801	[0.785, 0.815]
Held-out within-primitive (counting)	4	800	0.710	[0.680, 0.741]
Held-out cross-primitive similar (direction)	2	400	0.835	[0.798, 0.870]
Held-out cross-primitive dissimilar (letter ID)	2	400	0.863	[0.830, 0.898]

E Qwen replication

We replicate the three-mix ablation on QWEN2.5-VL-7B as a model-agnostic sanity check. Table 16 reports pass@ k at each mixture’s best-validation checkpoint, and Figure 11 shows the corresponding pass@ k curves against the untrained baseline.

Three observations complement the headline summary in §4.1:

- **Cross-mix pass@1 ordering is preserved** (invbase > uniform > cognitive) despite the smaller absolute gain, indicating the data-mix effect is not Apertus-specific.
- **Best step varies sharply with mixture**: uniform peaks at step 125, invbase at 280, cognitive at 310. The early peak for uniform is consistent with the entropy-collapse pattern we observe during training: with 13 of 22 task variants in counting, the GRPO group composition saturates fastest under uniform allocation.
- **The pass@ k trade-off is sharper**: all three RL variants lose 5–7 pp pass@128 relative to the baseline (vs. 2–3 pp for Apertus). This is consistent with Qwen’s stronger zero-shot prior leaving less prior-induced wrong mass for the contrastive signal to exploit.

Table 15: Pairwise pass@128 deltas for the held-out experiment. Confidence intervals are bootstrap 95% CIs over problems.

Comparison	Δ pass@128	95% CI
In-distribution – within-counting	+0.091	[+0.055, +0.126]
In-distribution – cross-similar	−0.034	[−0.073, +0.005]
In-distribution – cross-dissimilar	−0.062	[−0.097, −0.027]
Cross-similar – cross-dissimilar	−0.028	[−0.078, +0.023]
Within-counting – cross-similar	−0.125	[−0.175, −0.079]

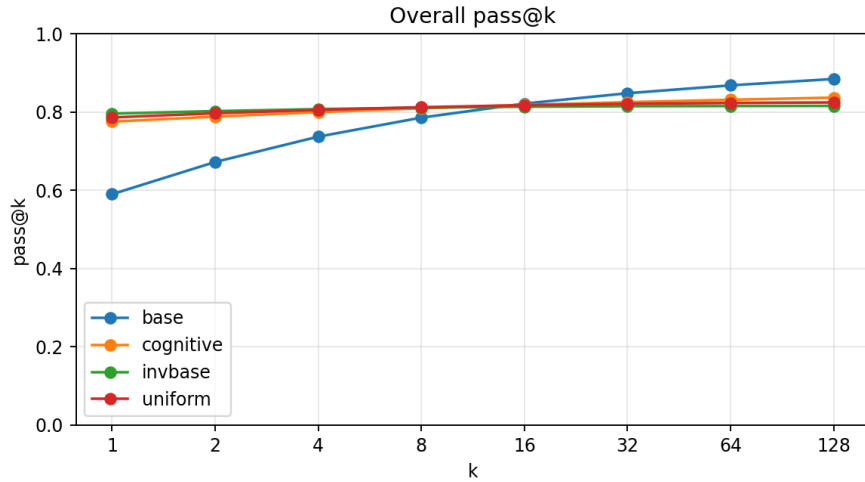


Figure 11: Pass@ k curves for the QWEN2.5-VL-7B baseline and the three RL-trained data mixtures at $T = 0.6$ with strict parsing. The base-vs-RL crossover happens between pass@8 and pass@32, earlier than the analogous Apertus crossover at pass@128 in Figure 6.

Table 16: Qwen2.5-VL-7B results for the three training mixtures.

Run	best step	pass@1	pass@8	pass@32	pass@128
Qwen base	—	0.590	0.786	0.848	0.885
Qwen + uniform	125	0.787	0.813	0.821	0.825
Qwen + cognitive	310	0.776	0.810	0.826	0.837
Qwen + inverse-baseline	280	0.796	0.812	0.815	0.816